

Supply & Demand

Unit 8 — Study Guide & Lecture Notes

- Price-taking firms: $MR = MC = P$ and the supply curve
- Competitive equilibrium, surplus & efficiency
- Shifts, elasticities, short-run vs long-run
- Taxes, incidence & deadweight loss

BEFORE YOU BEGIN

How to use these notes & where this unit fits

This guide follows the Unit 8 slides section-by-section and expands the reasoning so you can read it alone or annotate it in class. Every supply-and-demand diagram is redrawn cleanly, with equilibria, surplus areas and tax wedges computed from the actual functions so the geometry is exact.

DEFINITION BOX

The precise definition to reproduce in the exam.

WORKED EXAMPLE

Numbers carried through, matching the slide examples.

LISTEN FOR IN THE LECTURE

Points usually stressed verbally — leave room for your own notes.

CHECK YOURSELF

Quick questions to test recall.

Where this unit sits

The previous unit studied firms with **market power** — price **setters** who face a downward-sloping demand curve and whose outcomes are generally **not Pareto-efficient**. This unit studies the opposite polar case: **price-takers** in **competitive markets**. The two guiding questions:

- How does the behaviour of a **price-taking** firm differ from a price-setting one?
- **Can competition improve market outcomes?** (Spoiler: under strong assumptions, competitive equilibrium is Pareto-efficient — it exhausts all gains from trade.)

Outline

1. **A.** Introduction — market power vs price-taking
2. **B.** Demand and supply — WTP, WTA, the firm's supply curve, market supply, elasticity

3. **C.** Competitive equilibrium — market clearing, surplus and efficiency
4. **D.** Factors that affect equilibrium — shifts, elasticities, short vs long run
5. **E.** Applications — taxes, incidence and deadweight loss
6. **F.** Perfect competition — the assumptions behind the efficiency result

SECTION A

Introduction

Two kinds of firm sit at opposite ends of a spectrum:

PRICE SETTER (PREVIOUS UNIT)

A firm with **market power** can **set its own price**. It faces a downward-sloping demand curve, produces where $MR = MC$ with a **markup** over cost, and the outcome is **generally not Pareto-efficient** (some gains from trade go unrealised).

PRICE TAKER (THIS UNIT)

A firm in a market with **many firms selling identical products** cannot influence the price. It **takes the market price as given** and just decides **how much** to produce. Its supply depends purely on its **costs**.

The unit builds up the price-taking model, derives the market supply curve, finds the competitive equilibrium, and asks whether competition delivers an efficient outcome — then stress-tests that conclusion with shifts, elasticities and taxes.

LISTEN FOR IN THE LECTURE

Keep the contrast alive throughout: a price **setter** chooses a **point on a demand curve**; a price **taker** chooses a **quantity along its own cost curve** at a price it cannot change. Almost every result in this unit flows from that single difference.

SECTION B

Demand & Supply

THE TWO CURVES

- **Demand curve** — the total quantity all consumers together want to buy at each price. It represents the **willingness to pay (WTP)** of buyers.
- **Supply curve** — the total quantity all firms together would produce at each price. It represents the **willingness to accept (WTA)** of sellers.

Supply in a competitive market

When there are **many firms** selling **identical (homogeneous)** products:

- Firms are **price-takers**.
- The demand curve facing an **individual** firm is **horizontal** at the current market price (sell as much as you like at that price; charge a cent more and you sell nothing). Contrast this with the downward-sloping **market** demand curve.
- A price-taking firm **cannot benefit** from charging a different price and **cannot influence** the market price.
- Its supply therefore depends only on the **costs it incurs in production**.

The individual firm's supply — a bakery

A bakery can make **120 loaves at a constant marginal cost of €1.5**. It can produce more, but must pay **overtime**, so marginal cost jumps to **€2.6** beyond 120. The market price is **€2.35**.

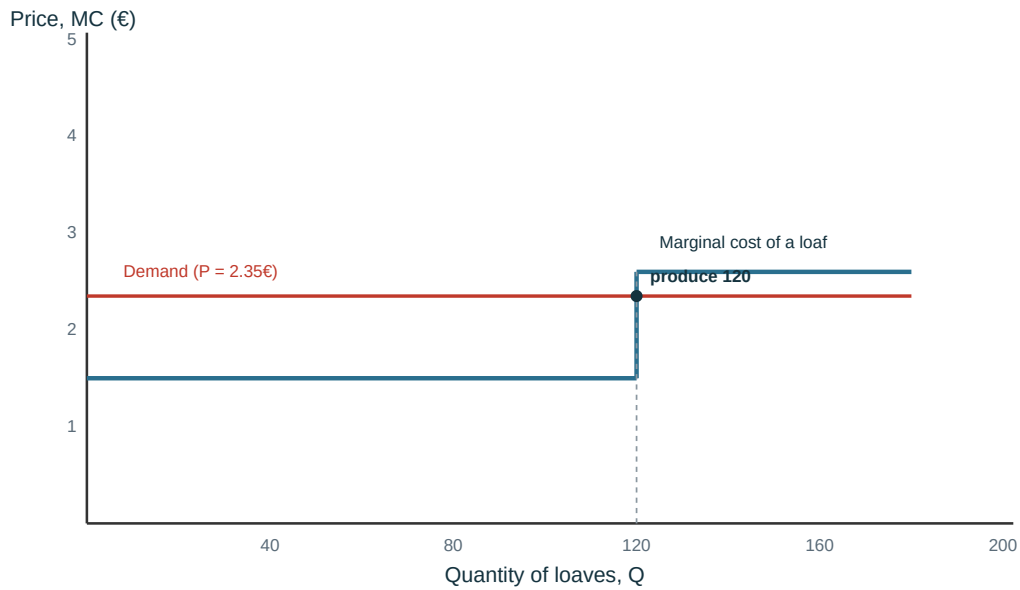


Figure 1. The firm's marginal cost is a **step**: $\text{€}1.5$ up to 120 loaves, then $\text{€}2.6$. The individual firm's demand is **horizontal** at the market price $\text{€}2.35$. **Rule: produce every unit whose price exceeds its marginal cost.** Here $\text{€}2.35 > \text{€}1.5$ but $< \text{€}2.6$, so the firm makes exactly **120 loaves**.

THE SUPPLY DECISION

- If **price** $>$ **marginal cost** of a unit $\rightarrow$ **produce and sell it**.
- If **marginal cost** $>$ **price** $\rightarrow$ **do not produce it**.
- Optimal production is where **price** $=$ **marginal cost**, so the firm's **supply curve is its marginal cost curve**.

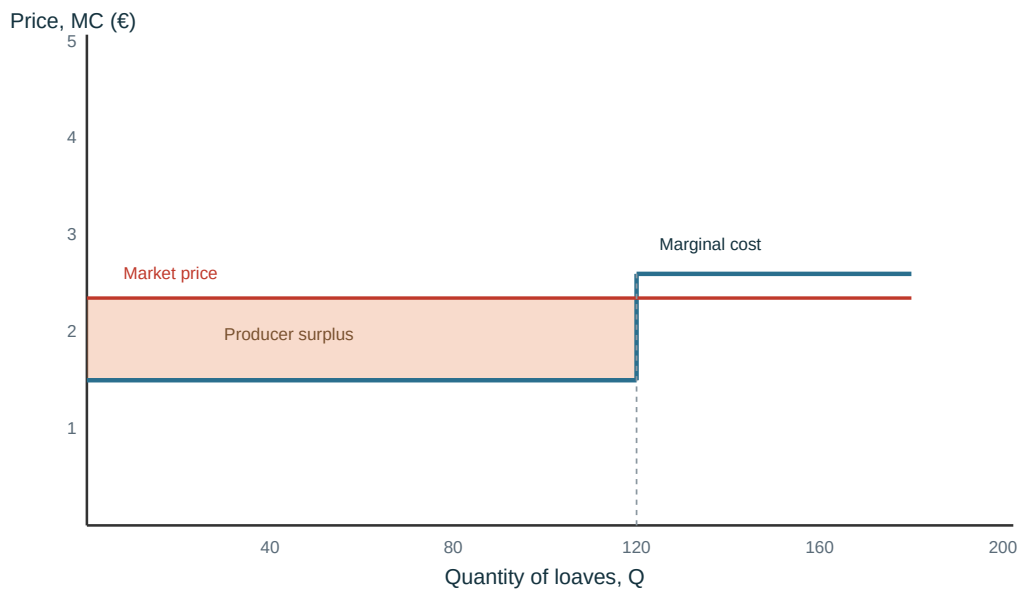


Figure 2. Producer surplus is the area between the market price and the marginal cost curve, over the units produced: here $(€2.35 - €1.5) \times 120 = \text{€}102$. It is the firm's gain from trading at the market price.

WHY $P = MC$ IS PROFIT-MAXIMISING ($MR = MC$)

Marginal revenue is the extra revenue from one more unit. The profit-maximising condition is $MR = MC$. For a **price-taker**, each unit sells at the market price, so $MR = P$. Hence the price-taker's rule $P = MC$ is exactly the general $MR = MC$ rule — perfectly consistent.

Should the firm produce at all? The shut-down check

Producing where $P = MC$ maximises profit *given* that you produce. But you must also check that revenue covers costs:

- Produce only if $P \geq AC$ (price at least covers **average cost**).
- Otherwise **exit** the market (unless it is a short-run phenomenon). So the supply curve is the part of the **MC curve that lies above the AC curve**.

From firm to market supply

The **market supply curve** is the **horizontal sum** of all firms' quantities at each price — add up how much every firm supplies at each price.

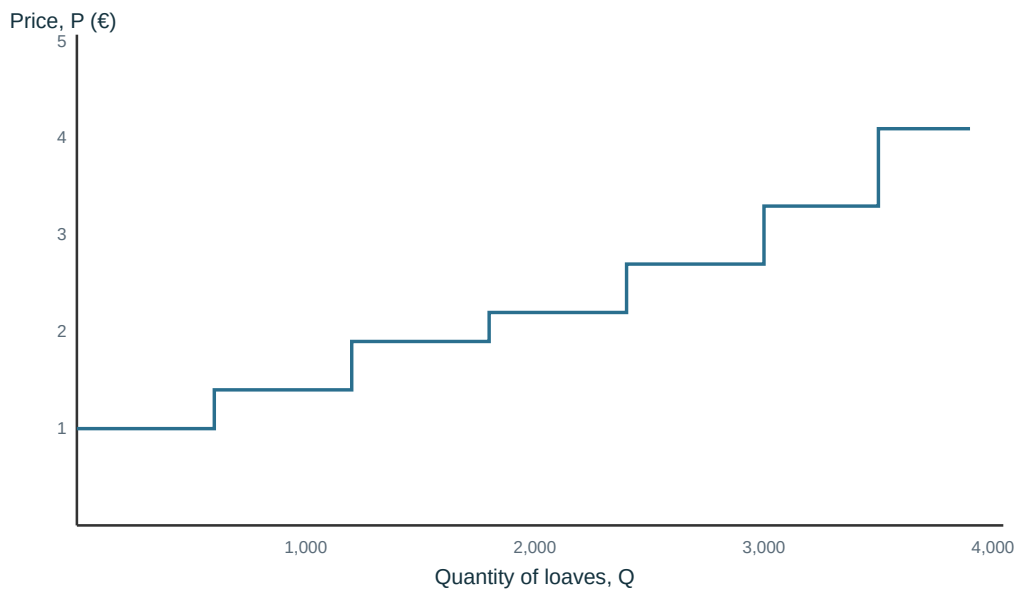


Figure 3. Summing many firms' step-shaped marginal costs produces a rising **staircase**: as price rises, more firms (and more of each firm's higher-cost units) become worth producing.

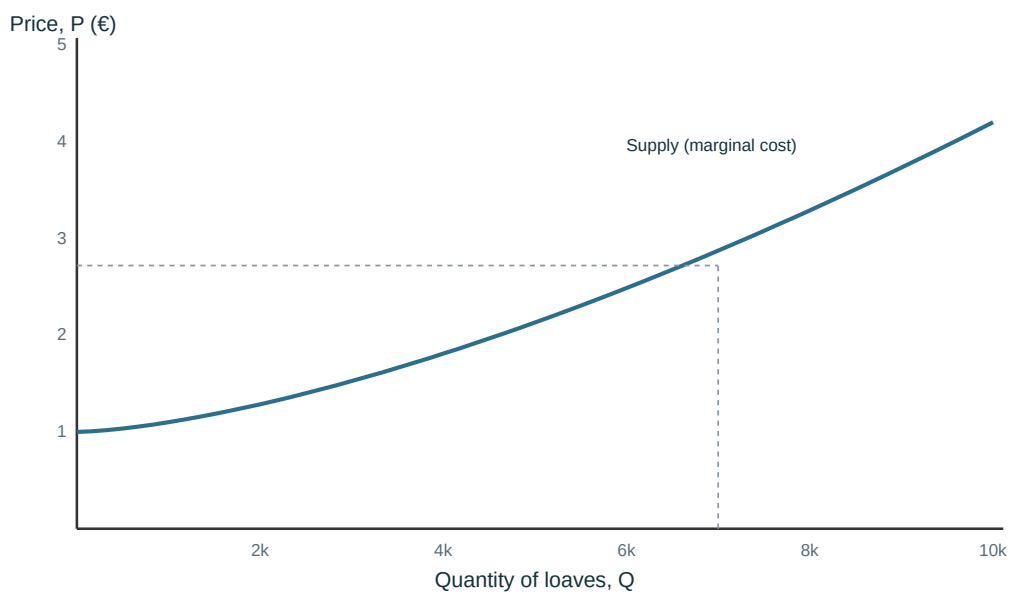


Figure 4. With many suppliers the staircase **smooths out** into an upward-sloping curve — the market's **marginal cost curve**. Higher prices call forth more output.

DEFINITION — PRICE ELASTICITY OF SUPPLY

How responsive quantity supplied is to price:

$$\epsilon_S = \frac{\% \Delta \text{ quantity supplied}}{\% \Delta \text{ price}}$$

Same characteristics as the elasticity of demand: a **flatter** supply curve is more **elastic** (very responsive), a **steeper** one is more **inelastic**. Elasticity will turn out to determine who bears a tax and how much prices move after a shock.

SECTION C

Competitive Equilibrium

Combine supply and demand to determine the equilibrium **price and quantity**.

DEFINITION — COMPETITIVE EQUILIBRIUM

A situation where **all buyers and sellers are price-takers** and, at the prevailing **market-clearing price P^*** , **quantity supplied = quantity demanded**.

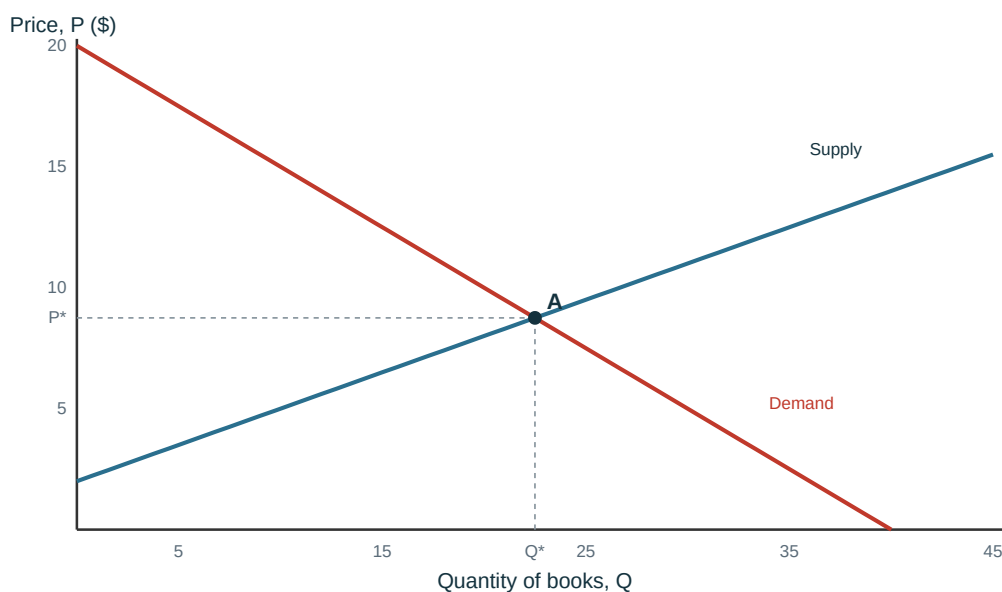


Figure 5. Equilibrium A sits where the supply and demand curves cross, pinning down the market-clearing price P^* and quantity Q^* .

Why the price is pushed to P^*

- If price is **above P^*** → **excess supply** (sellers want to sell more than buyers will buy) → **downward** pressure on price until $P = P^*$.
- If price is **below P^*** → **excess demand** → buyers **bid the price up** until $P = P^*$.
- Because products are **identical**, buyers will buy from any seller, and everyone **takes the prevailing price as given**.

Worked example — tanker (shipping) market

Demand and supply as functions of price p :

$$D = 40 - 0.15p \quad S = -10 + 0.10p$$

Market clears when $S = D$:

$$40 - 0.15p = -10 + 0.10p \implies 50 = 0.25p \implies p = 200, \quad D = S = 10$$

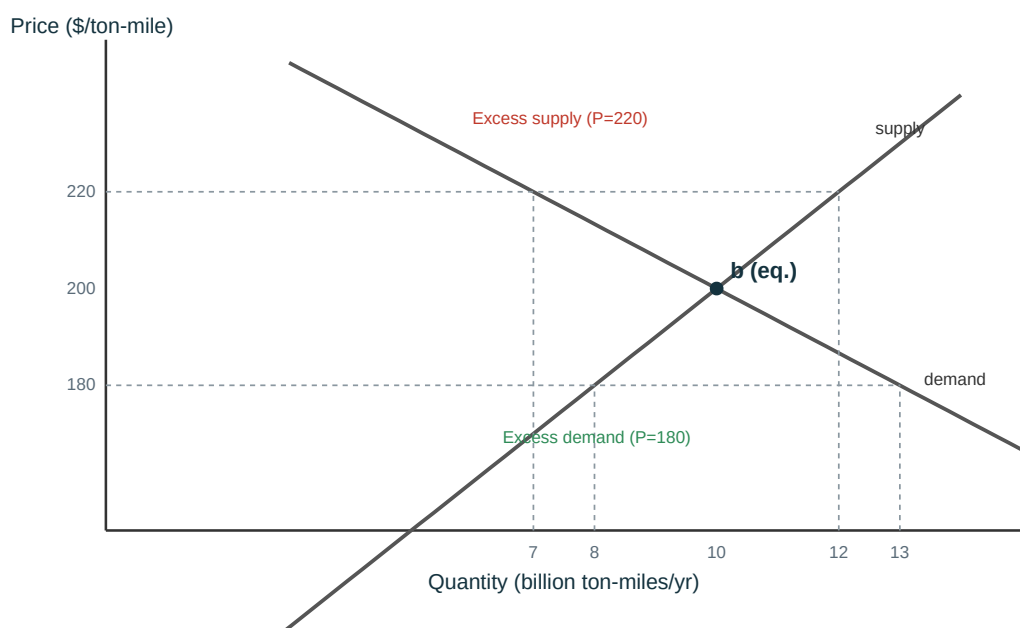


Figure 6. Equilibrium at $p = 200$, $q = 10$. At $p = 220$ quantity supplied (12) exceeds demanded (7) → **excess supply** → price pushed down. At $p = 180$ demanded (13) exceeds supplied (8) → **excess demand** → price pushed up. Both forces drive p back to 200.

Efficiency — all gains from trade are exploited

KEY EFFICIENCY RESULT

At the competitive equilibrium the **sum of consumer and producer surplus is maximised** — there is **no deadweight loss**. Every unit whose willingness to pay exceeds its marginal cost gets produced, and no unit beyond that. This is **not** the same as the point where *consumer* surplus alone is maximised (buyers would prefer a lower price, but that would shrink total surplus).

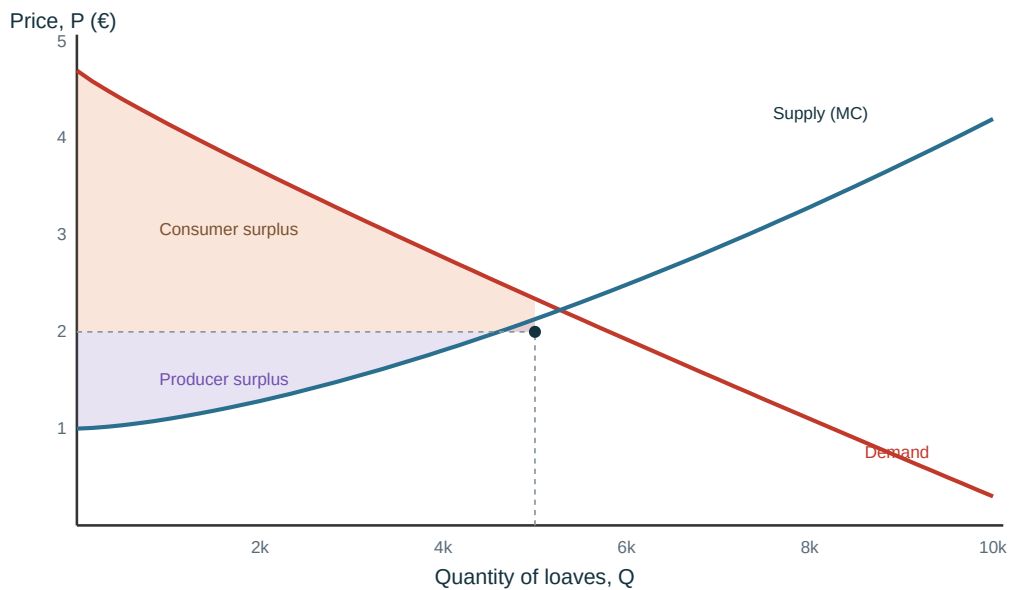


Figure 7. Consumer surplus (between demand and price) plus producer surplus (between price and supply) is as large as possible at equilibrium. Any other quantity would leave some mutually beneficial trades unmade — a deadweight loss.

When is competitive equilibrium Pareto-efficient? Assumptions & caveats

ASSUMPTIONS FOR EFFICIENCY

- Participants are **price-takers** (many firms, homogeneous products).
- Contracts are **complete**.
- Product quality/characteristics are **observed**.
- The transaction **only affects buyers and sellers** — **no external effects** (e.g. no pollution, cf. Unit 12).

CAVEATS

- If the assumptions fail, the allocation **may not be efficient**.
- **Fairness:** the split of total surplus depends on **elasticities** — the side that is **less elastic captures a smaller share** (share of surplus is inversely related to elasticity).
- Genuine price-takers are **hard to find** in real life.

CHECK YOURSELF

At the tanker equilibrium, what happens to price if a regulation fixes $p = 220$? Who is rationed — buyers or sellers? Why does the market-clearing price maximise *total* surplus but not *consumer* surplus?

SECTION D

Factors that Affect Equilibrium

The whole supply or demand curve can **shift** due to **exogenous shocks** — technological change, a change in popularity, input prices, entry/exit. Buyers and sellers then adjust until the market clears again at a **new equilibrium**.

A supply shift — improved baking technology

1. Better technology **lowers marginal costs** → the **supply curve shifts down/out**.
2. At the old price there is now **excess supply** → a **move along the demand curve**.
3. The price **falls** to a new, lower equilibrium (and quantity rises).

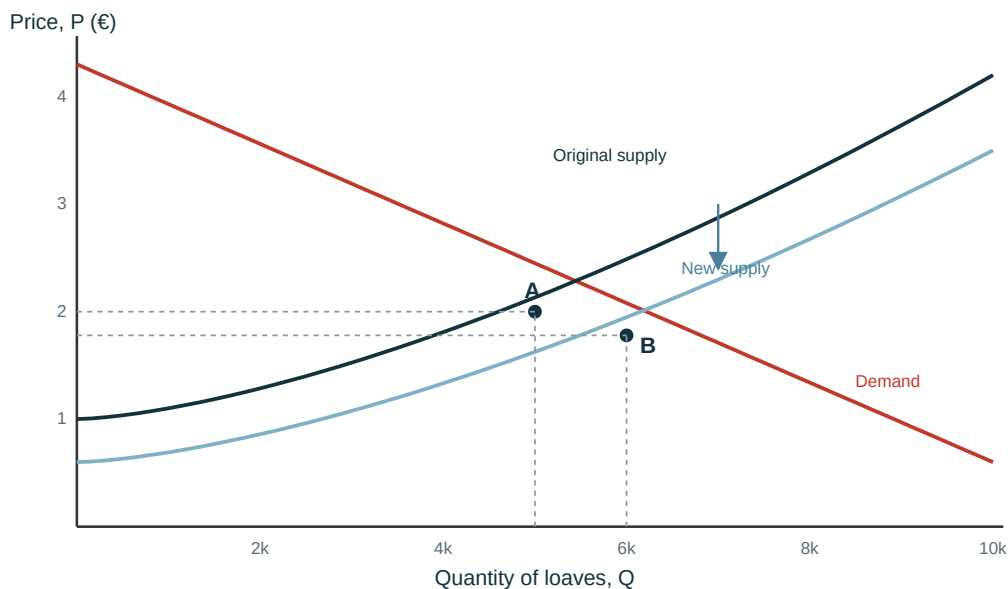


Figure 8. A cost-reducing shock shifts supply out (A → B): equilibrium price falls and quantity rises as the market re-clears along the demand curve.

COMMON MISCONCEPTION

People often expect the market price to fall by **the same amount** as the cost reduction. **Not so** — how much price falls (vs how much quantity rises) depends on the **price elasticities** of demand and supply. This is why manufacturers complain that retailers don't cut retail prices one-for-one with wholesale prices.

Why elasticity decides the price-vs-quantity split

Take the **same** downward supply shift, but with two different demand curves through the original equilibrium A:

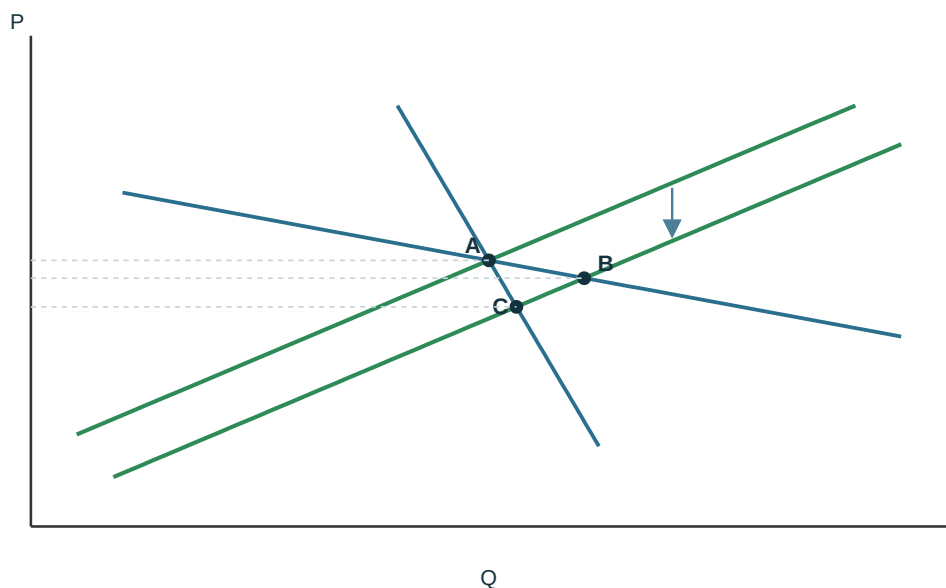


Figure 9. With **elastic** (flat) demand, the new equilibrium is **B**: price drops a little, quantity rises a lot. With **inelastic** (steep) demand, it is **C**: price drops a lot, quantity barely moves. Same cost shock, very different price response — set by demand elasticity.

Other shocks — entry and exit

- The supply curve also shifts through **market entry/exit**. If existing firms earn **economic rents** and the **costs of entry** are not too high, **new firms enter** → supply shifts **outward (down)**.
- **Demand shocks** are analysed the same way; their price impact depends on **supply** elasticity.

Short-run vs long-run equilibrium

In the bakery market, suppose equilibrium A has price €2. At that price an **individual bakery is making a profit** (you can see it because price sits **above its average cost**). That profit attracts entry.

LONG-RUN EQUILIBRIUM CONDITION

New bakeries keep entering (each wanting a share of the rents) → supply **shifts out and flattens** → market price and profits **fall** — until economic rent is **zero**:

$$P = MC = AC$$

At this long-run point (C) firms just break even and entry stops.

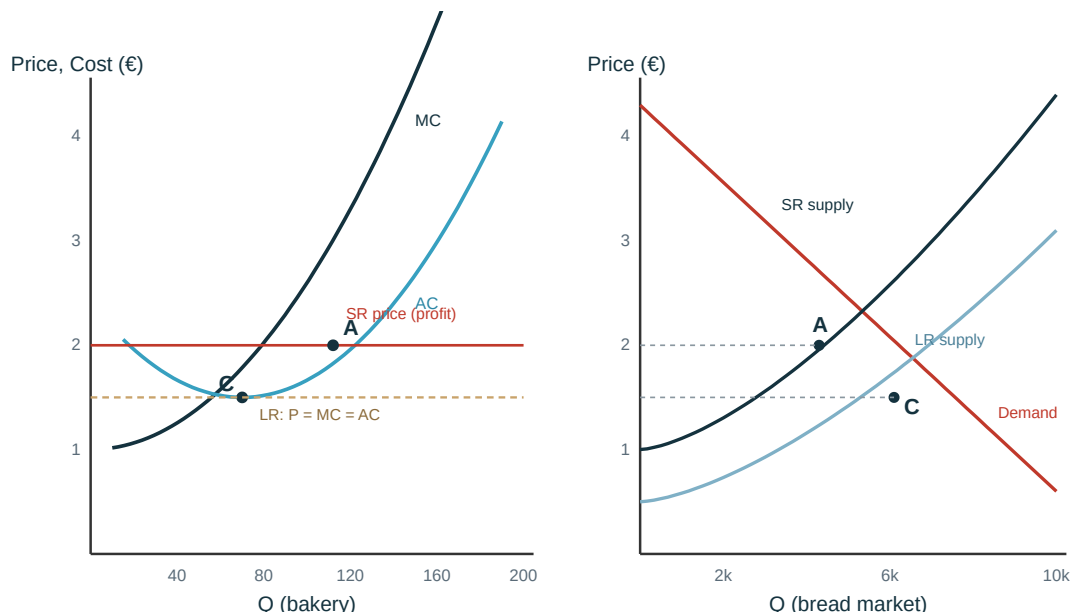


Figure 10. Left: the firm's MC and U-shaped AC. Short-run price (A) sits above minimum AC → economic profit. Right: that profit draws entry, shifting market supply out until price falls to minimum AC (C), where $P = MC = AC$ and rent is zero. The difference between short and long run depends on the production process (a gardener adjusts fast; a chemical plant slowly).

SHORT-RUN VS LONG-RUN ELASTICITY

The number of firms is **exogenous in the short run** but **endogenous in the long run**. Therefore **supply is more elastic in the long run**, as more firms can enter. **Investment warning:** a positive demand shock raises prices, but the rise is **smaller in the long run**; basing an investment on the short-run price spike risks **over-investment**.

SECTION E

Applications — Taxes

Governments tax to **raise revenue** and/or **discourage consumption**. Two types:

TWO KINDS OF TAX

- **Ad valorem tax** — a fixed **percentage of value** (e.g. VAT).
- **Specific (unit) tax** — a fixed **amount per quantity** (e.g. €10 per airline ticket).

A tax can be collected from **either side**, and it drives a **wedge** between three prices:

- **Seller price p_S** — what the seller keeps after paying the tax.
- **Buyer price p_B** — what the buyer actually pays, tax included.
- **Market price p** — the price at which the good trades.

For a specific tax t : if the **seller** remits it, $p_S = p - t$ and $p_B = p$. If the **buyer** remits it, $p_B = p + t$ and $p_S = p$. In both cases: **express demand and supply as functions of the market price, then set supply = demand**.

Worked example — a €10 airline-ticket tax

Demand and supply (thousands of tickets):

$$q_D = 4920 - 5 \cdot p_B \quad q_S = -1746.67 + \frac{10}{3} \cdot p_S$$

No tax: buyer price = seller price = market price → equilibrium **b** at **p = 800, q = 920**.

Tax of €10 collected from sellers ($p_S = p - 10$): supply becomes $q_S = -1746.67 + \frac{10}{3}(p - 10) = -1780 + \frac{10}{3}p$. Setting $q_D = q_S$ gives market price **p = 804** (= the buyer price), quantity **q = 900**, and seller price **$p_S = 804 - 10 = 794$** .

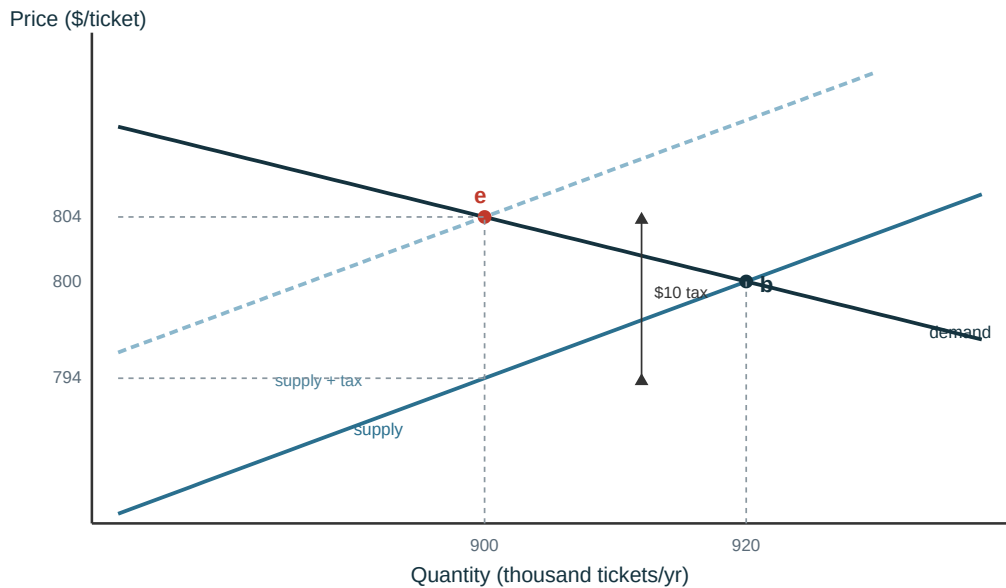


Figure 11. Collecting the tax from sellers shifts the supply curve **up by €10** (sellers now need €10 more per ticket). Equilibrium moves from **b** (800, 920) to **e** (804, 900). The tax drives a **wedge**: buyers pay €804, sellers keep €794.

TAX EQUIVALENCE — WHO "PAYS" DOESN'T MATTER

Collect the **same €10 from buyers instead** ($p_B = p + 10$): demand becomes $q_D = 4920 - 5(p + 10) = 4870 - 5p$. Equating with the untaxed supply gives market price **$p = 794$** , buyer price **$794 + 10 = 804$** , seller price **794**, quantity **900** — **exactly the same outcome**. Whether the buyer or the seller physically remits the tax is irrelevant to prices and quantities. **Subsidies are just negative taxes** and are analysed the same way (shift supply down or demand up).

Surplus, government revenue and deadweight loss

Before the tax, total welfare = consumer surplus + producer surplus. The tax rearranges this:

- **Consumer surplus** — area under demand above the **buyer** price (804).
- **Producer surplus** — area above supply below the **seller** price (794).
- **Government revenue** — the tax $\times$ quantity traded (10×900).
- **Total welfare = CS + PS + government revenue.**

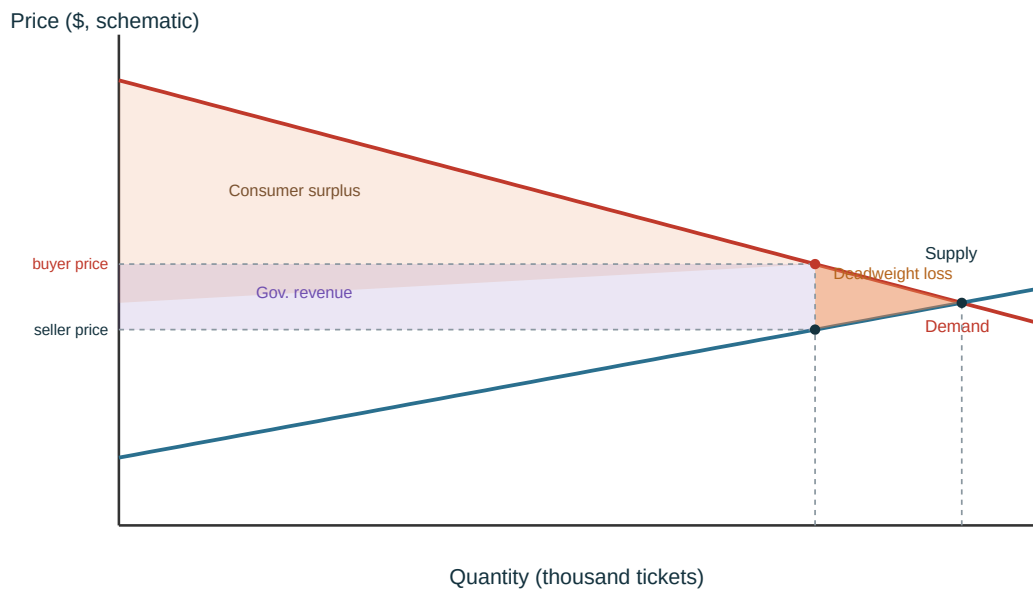


Figure 12. The tax splits the price into buyer (804) and seller (794) prices and cuts quantity from 920 to 900. Government collects the middle rectangle. But the **orange triangle** — trades that were worthwhile ($WTP > MC$) yet no longer happen — is lost to everyone: the **deadweight loss**. Total welfare falls by this triangle.

WHY DEADWEIGHT LOSS ARISES

The tax drives a wedge between the **marginal cost of producing** and the **marginal benefit to consumers**. Some consumers whose willingness to pay **exceeds** the cost of production can no longer buy → total surplus falls and quantity traded is **too low**.

Tax incidence — who really bears the tax?

INCIDENCE & ELASTICITY

The effect of a tax is the **same whether collected from buyers or sellers** (equivalence). With moderate elasticities: the **buyer price rises by less than the tax**, the **seller price falls by less than the tax**, and the **quantity falls**. Crucially, **the side of the market that is less sensitive to price (less elastic) bears the larger share of the tax**.

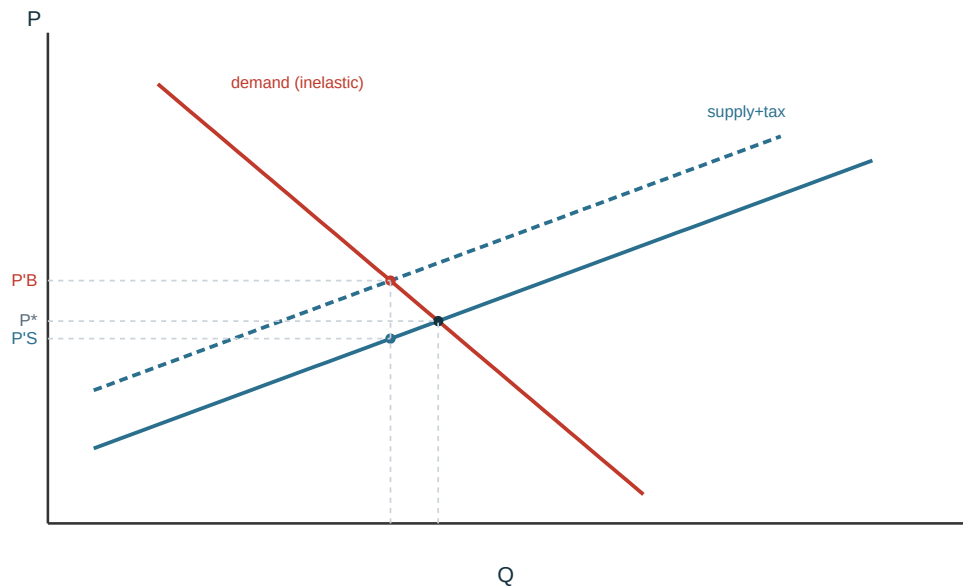


Figure 13. When **demand is more inelastic than supply**, buyers bear more: the buyer price jumps from P^* up to P'_B (a lot), while the seller price only slips to P'_S (a little). In the airline example demand is more elastic than supply, so the split is the other way — the buyer price rises €4 and the seller price falls €6.

Real-world example — Denmark's butter (saturated-fat) tax

In 2011 Denmark introduced a tax per kg on saturated fat, **≈ 22% of the average butter price**. Consumption of butter and related products **fell 15–20%**. But the tax was **scrapped** soon after because of the **administrative burden** of collecting it.

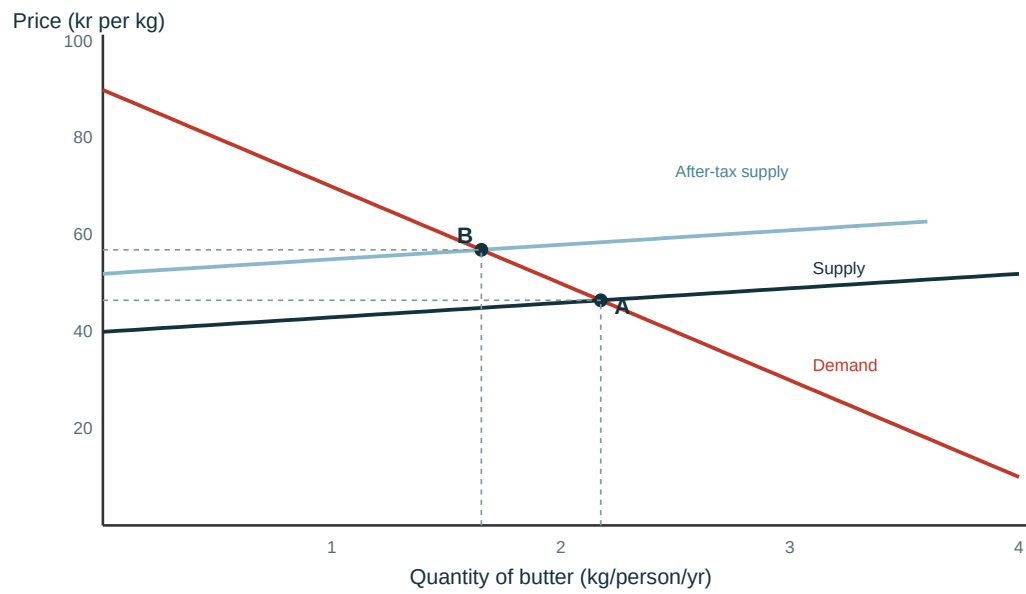


Figure 14. The saturated-fat tax shifts the supply curve up; equilibrium moves from A to B — a higher consumer price and lower quantity. It worked (consumption fell) but the collection costs outweighed the benefits, and it was repealed.

SECTION F

Perfect Competition

The efficiency result of Section C is the pay-off of a specific idealised market structure, **perfect competition**. It pulls together every assumption used along the way.

THE PERFECTLY COMPETITIVE MODEL

A market is **perfectly competitive** when:

- there are **many buyers and sellers**, all **price-takers**;
- products are **homogeneous** (identical);
- there is **free entry and exit** in the long run;
- information is good and there are **no external effects**.

Under these conditions the market reaches $P = MC$ in the short run and $P = MC = AC$ in the long run, and the equilibrium is **Pareto-efficient** — it exhausts all gains from trade with **no deadweight loss**.

LISTEN FOR IN THE LECTURE

Perfect competition is a **benchmark**, not a description of reality ("hard to find price-takers in real life"). Its value is as a yardstick: departures from it — market power (previous unit), taxes (Section E), and externalities (Unit 12) — are exactly the cases where the efficiency result **breaks**, and each creates a deadweight loss.

CHECK YOURSELF

- Which single assumption fails for a firm with market power, and why does that create deadweight loss?
- In long-run equilibrium a competitive firm earns zero economic profit — is it therefore making an accounting loss? (Recall Unit 2: economic vs accounting profit.)

Summary, Formulas & Glossary

The unit in one paragraph

In a competitive market, firms are **price-takers**: each faces a horizontal demand at the market price and produces where $P = MC$ (so its supply curve is its marginal-cost curve above AC). Summing firms gives the upward-sloping **market supply**. The **competitive equilibrium**, where supply = demand, **maximises total surplus** (CS + PS) with **no deadweight loss** — provided price-taking, complete contracts and no externalities hold. Shocks **shift** the curves; how price vs quantity responds depends on **elasticities**, and **entry** drives the long run to $P = MC = AC$ (zero rent). A **tax** drives a wedge between buyer and seller prices, is **independent of who remits it**, transfers surplus to government, and creates a **deadweight loss**; its **incidence** falls more heavily on the **less elastic** side.

Formula & rule sheet

Concept	Rule / expression
Price-taker's output	produce where $P = MC$ (since $MR = P$)
Firm supply curve	the MC curve, for the part above AC
Shut-down / exit rule	produce only if $P \geq AC$ (long run)
Producer surplus	area between price and MC, over units sold
Market supply	horizontal sum of firms' quantities at each price
Elasticity of supply	$\epsilon_S = \% \Delta \text{ quantity supplied} / \% \Delta \text{ price}$
Competitive equilibrium	quantity supplied = quantity demanded ($P = P^*$)
Long-run equilibrium	$P = MC = AC$ (zero economic rent)
Specific tax (seller remits)	$p_S = p - t$, $p_B = p$
Specific tax (buyer remits)	$p_B = p + t$, $p_S = p$
Total welfare with tax	CS + PS + government revenue (– deadweight loss vs no tax)
Tax incidence	less elastic side bears the larger share

Glossary

Term	Meaning
Price-taker	Firm/agent that takes the market price as given; faces horizontal demand.
WTP / WTA	Willingness to pay (demand) / willingness to accept (supply).
Marginal cost / revenue	Cost / revenue of producing one more unit; optimum at $MR = MC$ (= P for a price-taker).
Producer / consumer surplus	Seller / buyer gains from trade (areas relative to price).
Elasticity of supply/demand	$\% \Delta \text{ quantity} \div \% \Delta \text{ price}$; flatter = more elastic.
Competitive equilibrium	Price-taking market where supply = demand.
Pareto efficiency	No one can be made better off without making another worse off; competitive eq. achieves it under the assumptions.
Deadweight loss	Lost total surplus from trades that don't happen (e.g. under a tax).
Short vs long run	Number of firms fixed vs free to enter/exit; long-run supply is more elastic.
Tax wedge / incidence	Gap between buyer and seller price / how the burden is split.
Ad valorem vs specific tax	% of value (VAT) vs fixed amount per unit.

Exam-style questions to practise

1. A firm has $MC = 2 + 0.01Q$. At a market price of €5, how much does it supply, and what is its producer surplus?
2. Demand is $Q = 100 - 2P$, supply $Q = -20 + 2P$. Find the equilibrium price and quantity, and identify excess supply/demand at $P = 35$ and $P = 25$.
3. A €4 specific tax is levied on a good. Explain why the outcome is identical whether buyers or sellers remit it.
4. Using elasticities, explain who bears more of a tax on cigarettes (inelastic demand) versus a tax on a specific brand of soft drink (elastic demand).
5. Draw the short-run and long-run adjustment to a permanent rise in demand for bread, and explain why the long-run price rise is smaller.
6. Why does a competitive equilibrium maximise *total* surplus but not *consumer* surplus? Illustrate the deadweight loss created by a price ceiling.

Study guide prepared as a companion to *Microeconomics — Prosperity and Inequality, Unit 8: Supply and Demand — Price-taking and Competitive Markets* (KU Leuven, 2025–2026). Equilibria, surplus areas and tax wedges are computed from the stated demand/supply functions; the airline and tanker examples reproduce the slides' numbers.